

HEP Significance Workflow

Introduction

This presentation summarizes the workflow and outputs of the HEP significance analysis for the SOLAR project.

- This deck is auto-generated from workflow outputs.
- This deck is scoped to the **Truncated** folder for the **SolarEnergy** reconstruction algorithm.

Config aliases:

- hd_1x2x6_centralAPA: HD Central
- hd_1x2x6_lateralAPA: HD Lateral
- vd_1x8x14_3view_30deg_nominal: VD Top
- vd_1x8x14_3view_30deg_shielded: VD Bottom Shielded

Workflow

- config: list of detector configs
- folder: **Truncated**
- analysis: HEP
- exposure: default **30 years**
- threshold in 13HEP.py: default 10.0 MeV
- optional cuts override: nhits, ophits, adjcls
- significance reference in plots: ProfileLikelihood
- best-curve reference in 0ZBestSigmas.py: Smoothed or Raw

Workflow Outputs

- Fiducial optimization: `data/solar/fiducial/truncated/BestFiducials.json`
- Best cut summaries (JSON): `data/analysis/daynight-json/truncated`
- Significance scans (PNFS outputs):
`/pnfs/ciemmat.es/data/neutrinos/DUNE/SOLAR/HEP/truncated`
- Figures: `images/analysis/hep/truncated`

Histogram and Significance Flow

- Step 1: Build HEP rates and threshold region in `src/analysis/13HEP.py` per component.
- Step 2: Apply component-aware smoothing via `lib/lib_smooth.py` using HEP smoothing config.
- Step 3: Evaluate Gaussian and Asimov significance curves in `src/analysis/13HEP.py`.
- Step 4: Pick best cuts in `src/analysis/0ZBestSigmas.py`, then evaluate ProfileLikelihood exposure in `src/analysis/13HEPProfileLikelihood.py`.
- Step 5: Render exposure/significance and comparison plots in `src/analysis/13HEPExposurePlot.py`, `src/analysis/13HEPSignificancePlot.py`, `src/analysis/13HEPSignificanceComparisonPlot.py`, and `src/analysis/13HEPExposureComparisonPlot.py`.

Adaptative Rebinning: Strategy

- Rebinning is applied in `src/analysis/13HEP.py` through `lib/lib_smooth.py` using `apply_adaptive_tail_rebin` .
- It is controlled by `import/analysis.json` under `ADAPTIVE_REBIN -> ANALYSES -> HEP` .
- At each exposure, bins are merged from the high-energy tail until the expected detectable signal per rebinned group reaches the configured threshold.
- This stabilizes low-statistics significance estimates while preserving discovery sensitivity in sparse tails.

Rebin threshold criterion used for each grouped bin:

$$S_{\text{group}} = \sum_{i \in \text{group}} S_i^{\text{det}} \geq T$$

$$T = \max(\text{min_expected_events}, -\ln(1 - \text{min_count_probability}))$$

Adaptative Rebinning: Discovery

Discovery significance is then evaluated on rebinned inputs:

$$Z = Z(S_{\text{group}}, B_{\text{group}}, \sigma_{B,\text{group}})$$

Latest ProfileLikelihood updates used in this deck:

- Adaptive rebinning is enabled by default in [src/analysis/13HEPProfileLikelihood.py](#).
- Detection-mask zeroing is disabled for the ProfileLikelihood path to avoid artificial early-exposure suppression.
- Adaptive rebin starts are frozen (computed once at reference exposure and reused across the scan) to reduce discontinuities and keep curves monotonic.

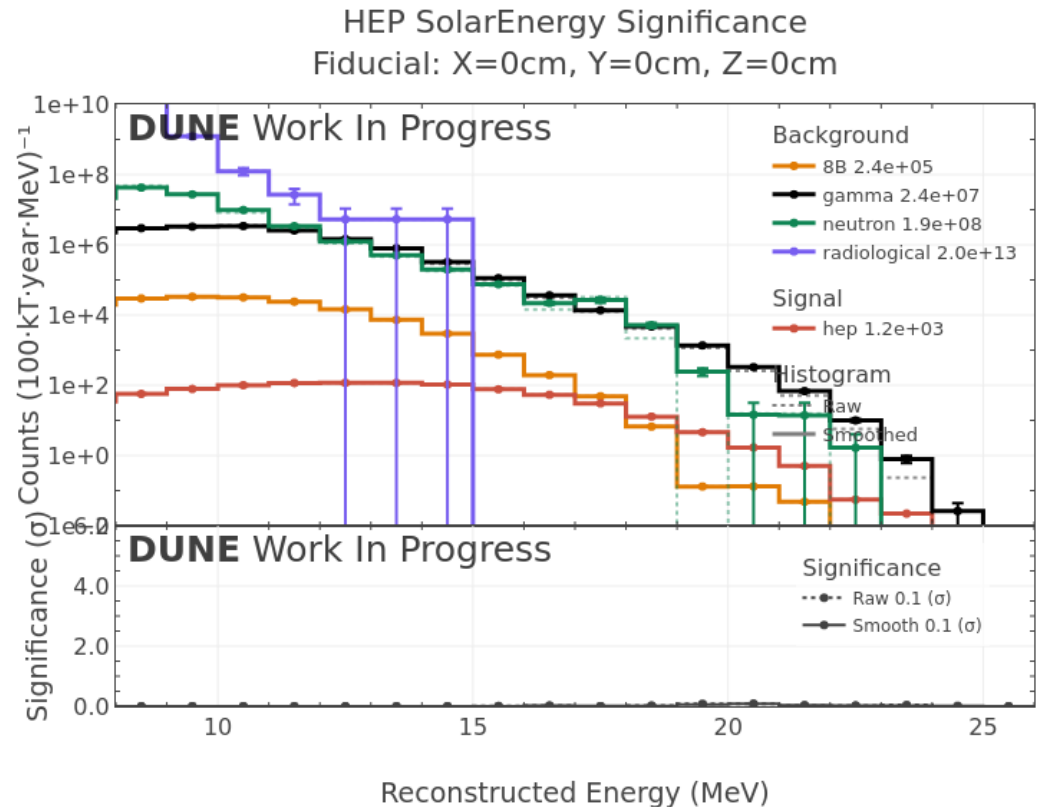
Smoothing by Stage (HEP)

Stage	Enabled	Method	Component Mode	Smoothed Components	Sigma
Fiducial	Yes	gaussian	only	gamma, neutron	0.50
Significance	Yes	gaussian	only	gamma, neutron	0.50

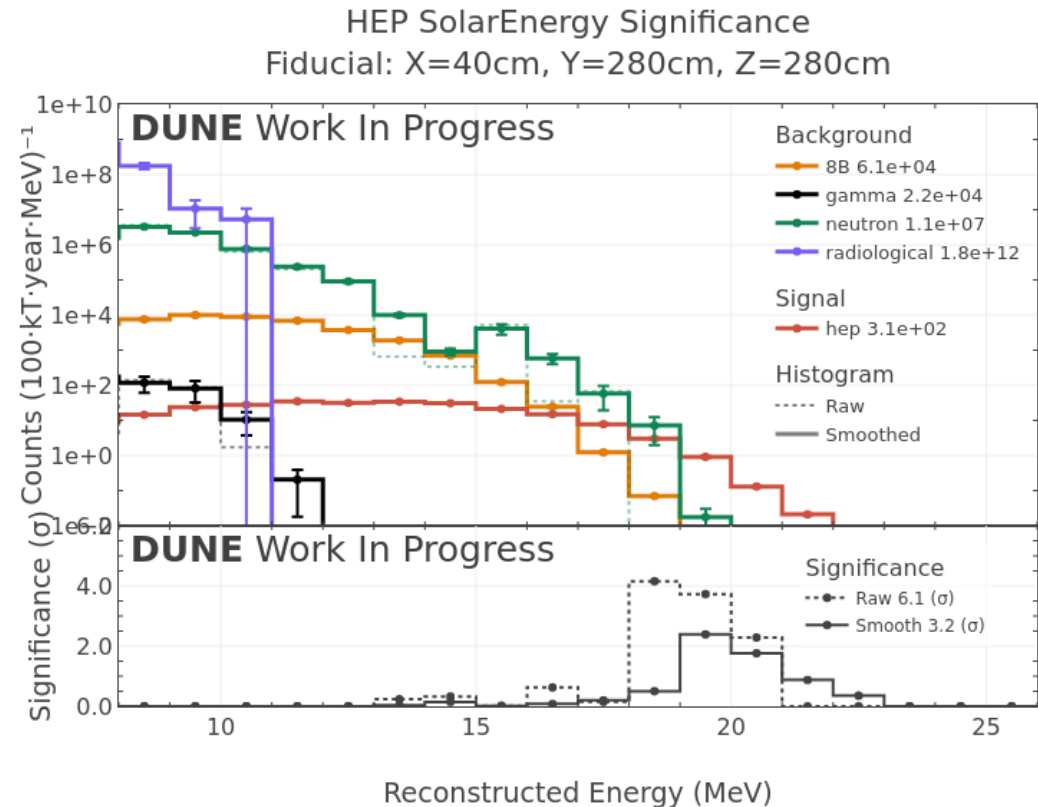
Fiducialization

HD Central

No Fiducial

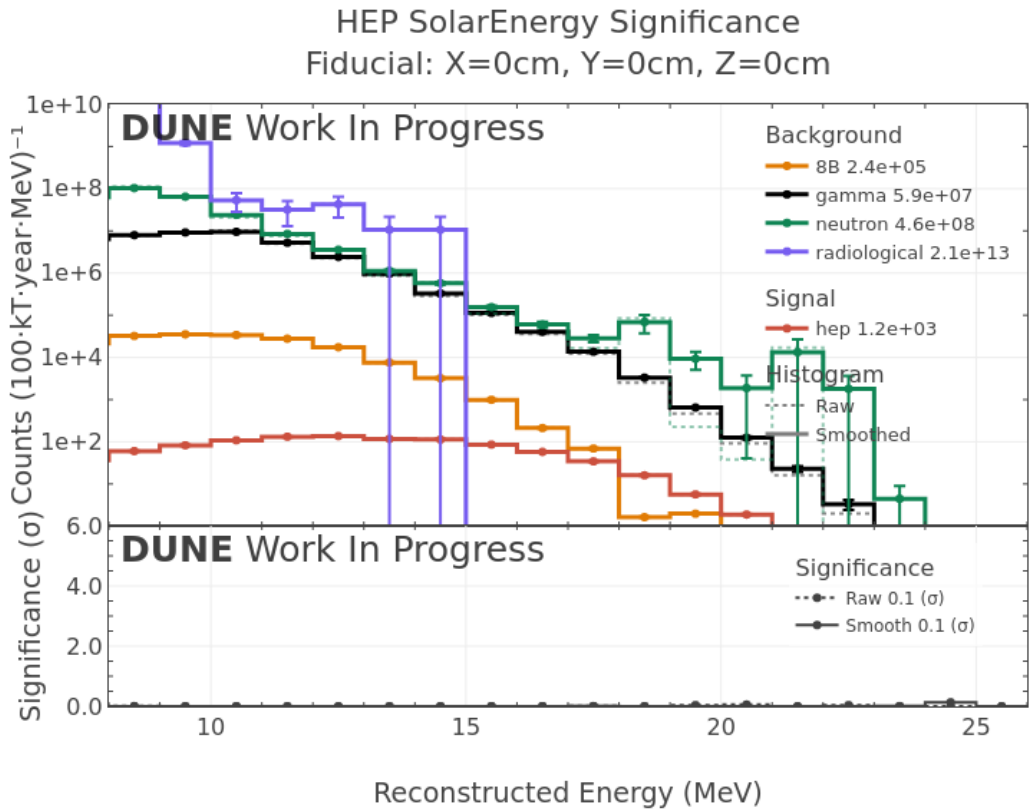


Best Fiducial

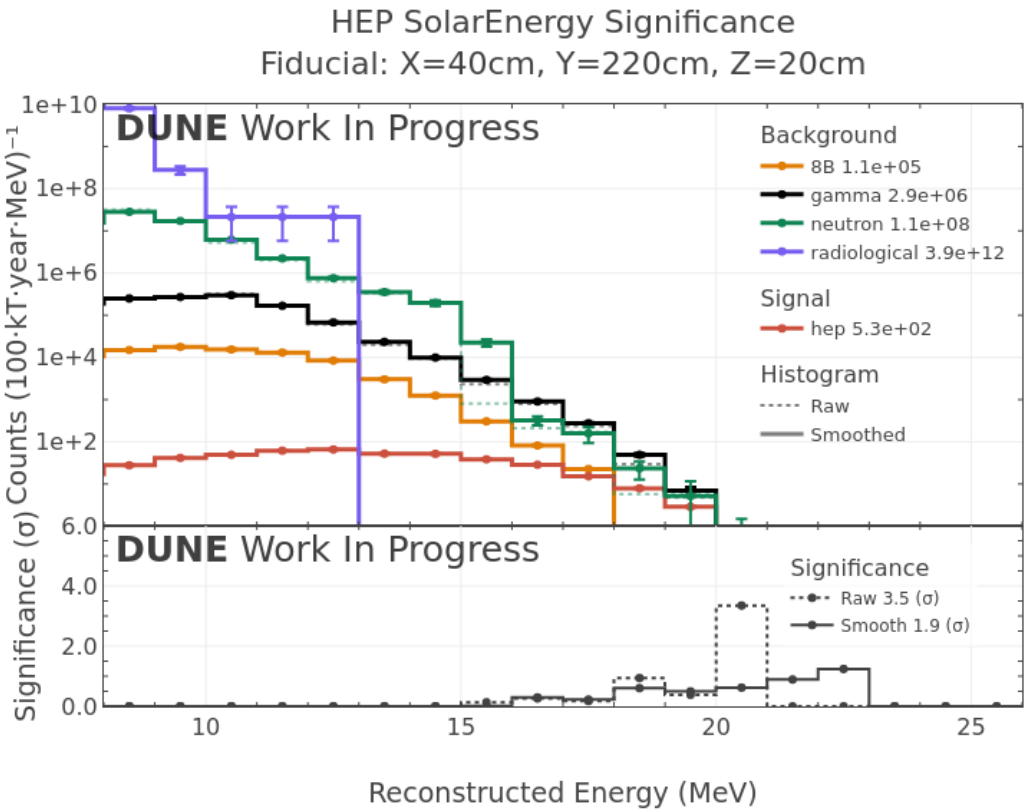


HD Lateral

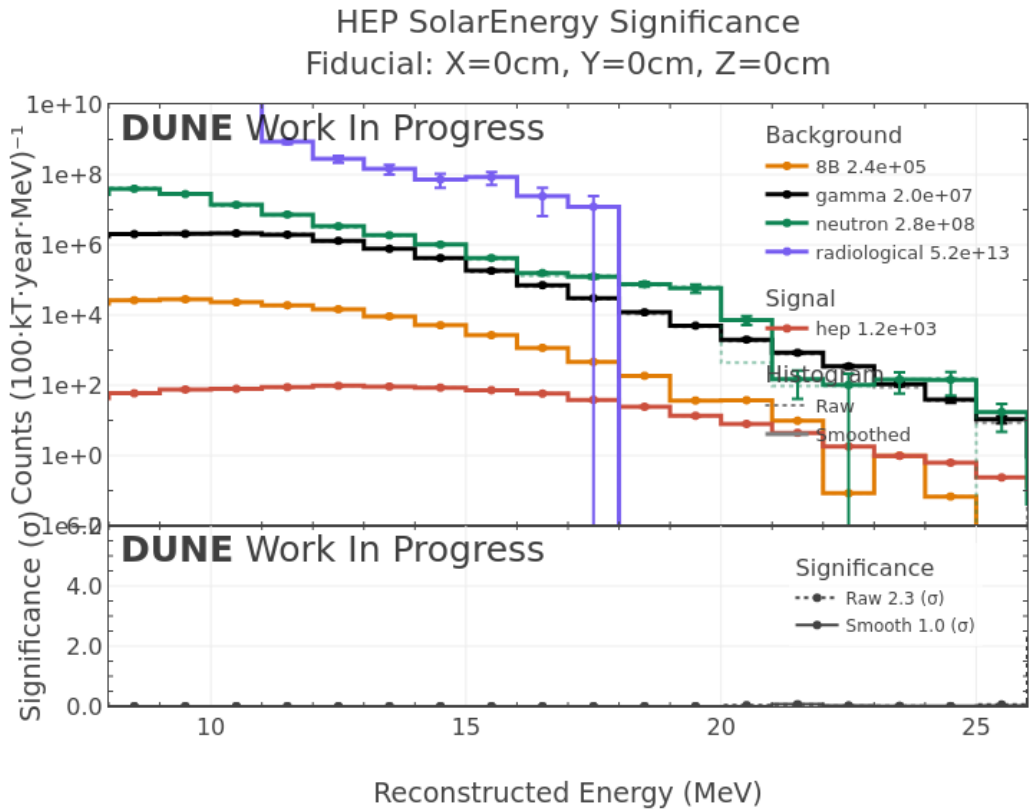
No Fiducial



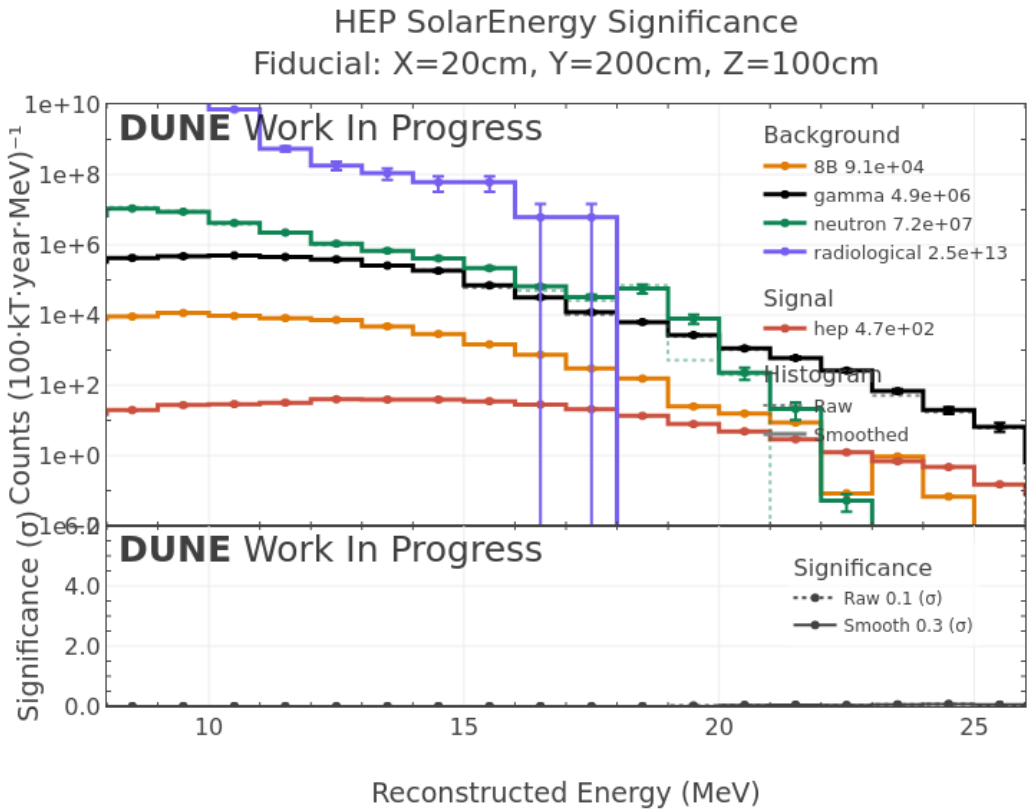
Best Fiducial



No Fiducial

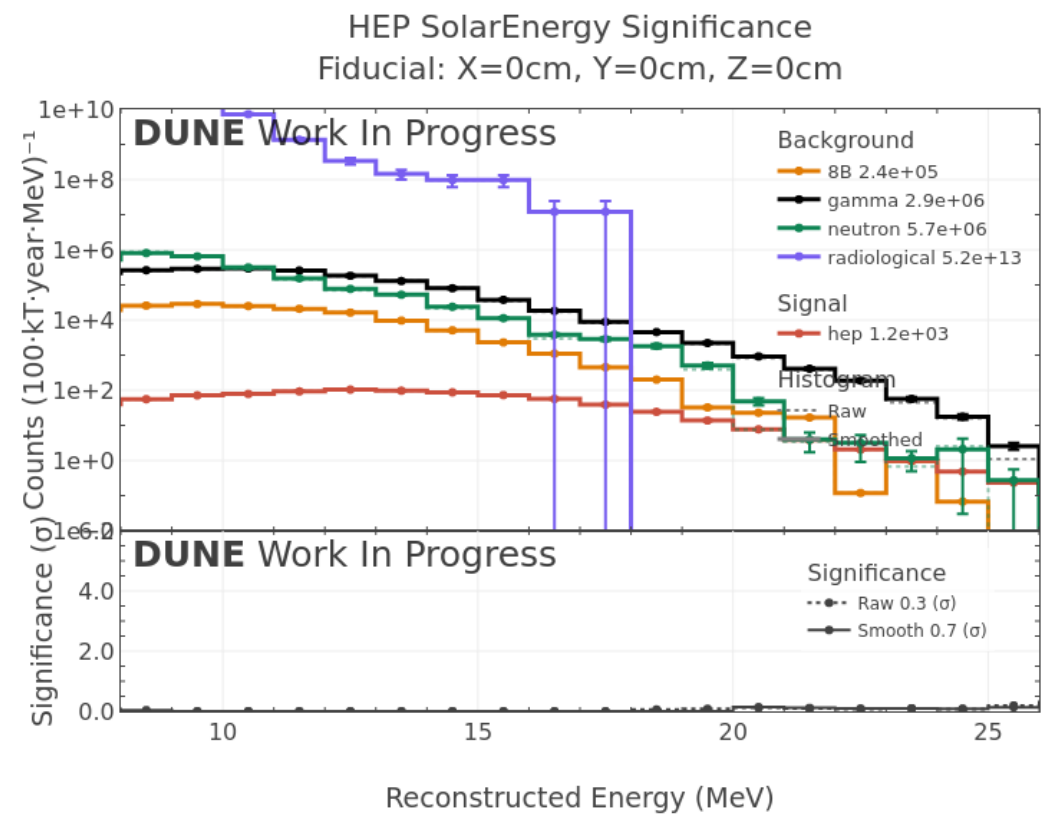


Best Fiducial

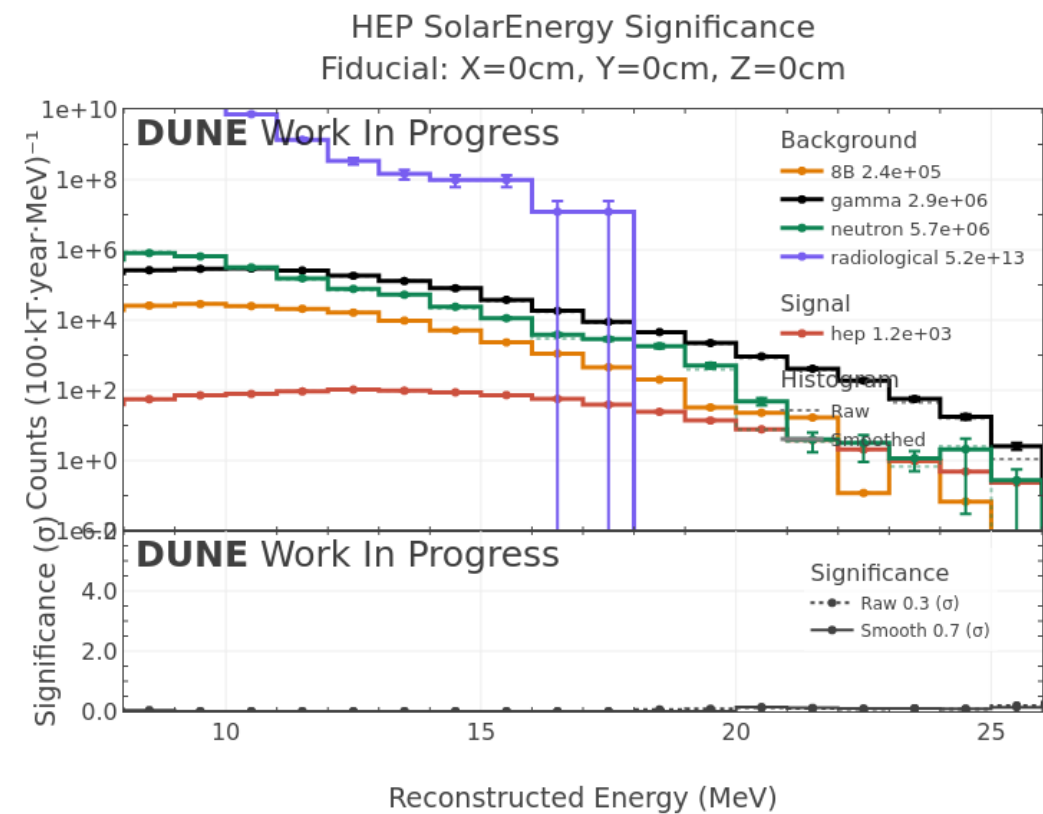


VD Bottom Shielded

No Fiducial



Best Fiducial



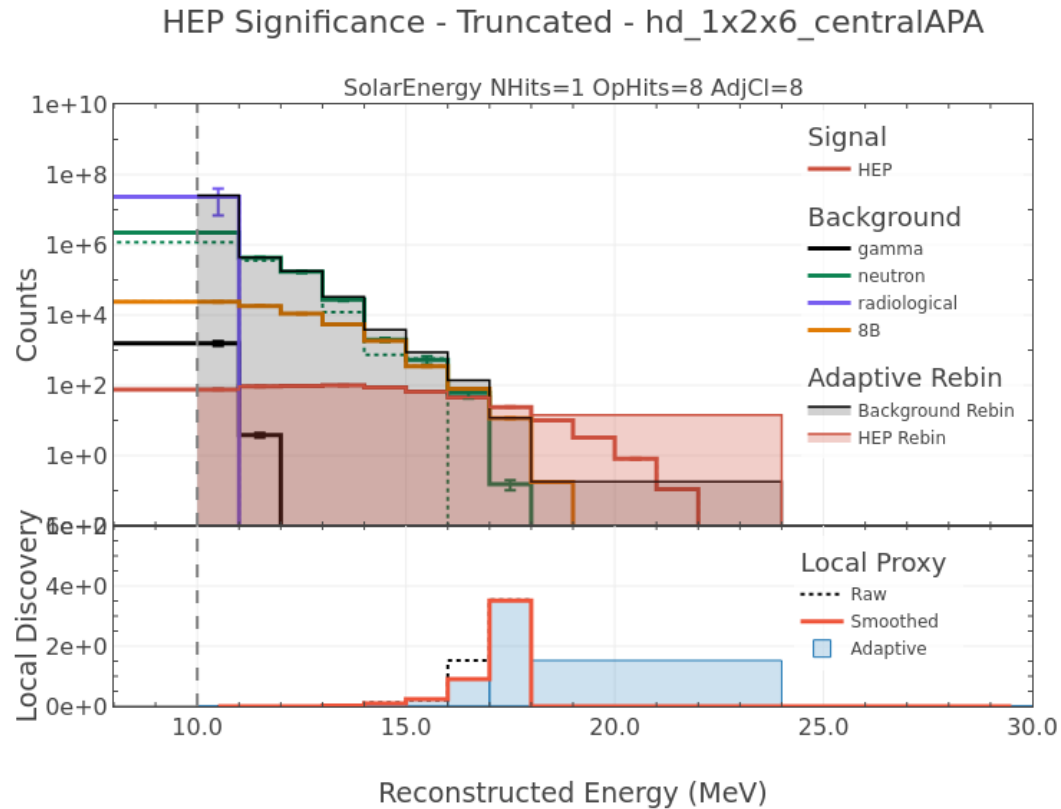
Fiducial Optimization Summary

Config	Fiducial X	Fiducial Y	Fiducial Z	Before Fiducialization	After Fiducialization
HD Central	40	280	280	0.091	1.789
HD Lateral	40	220	20	0.078	1.132
VD Top	20	200	100	0.072	0.110
VD Bottom Shielded	0	0	0	0.393	0.393

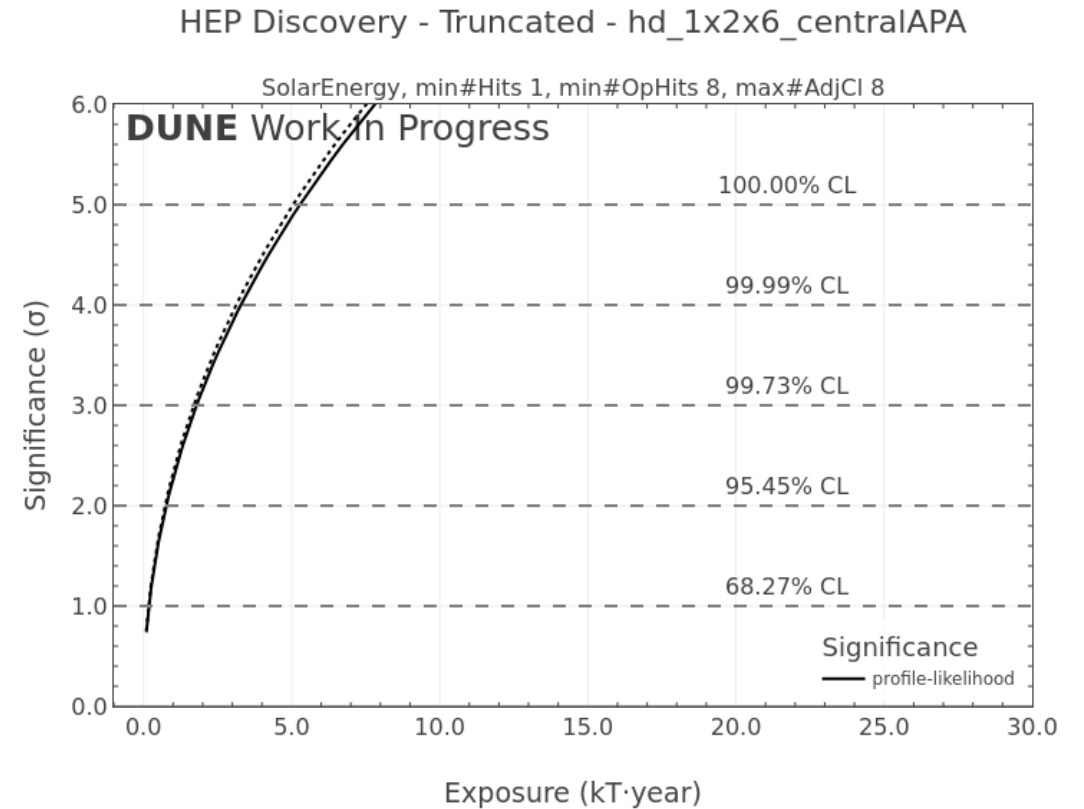
HEP Results

HD Central

Significance



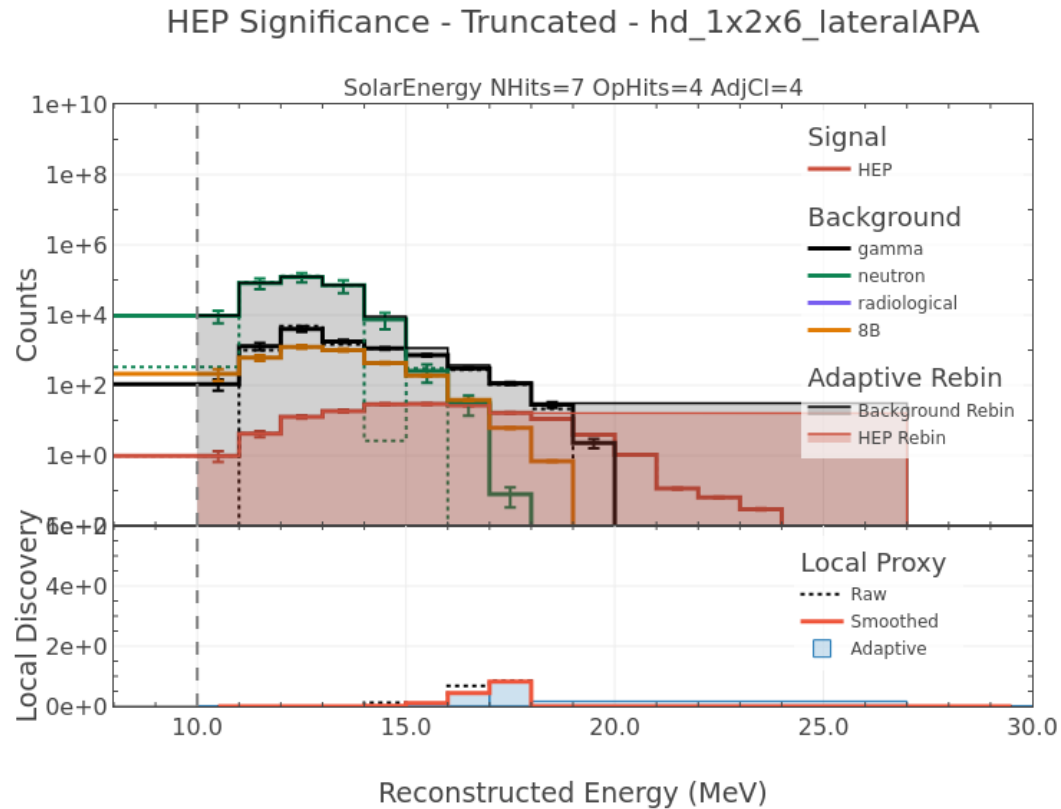
Exposure



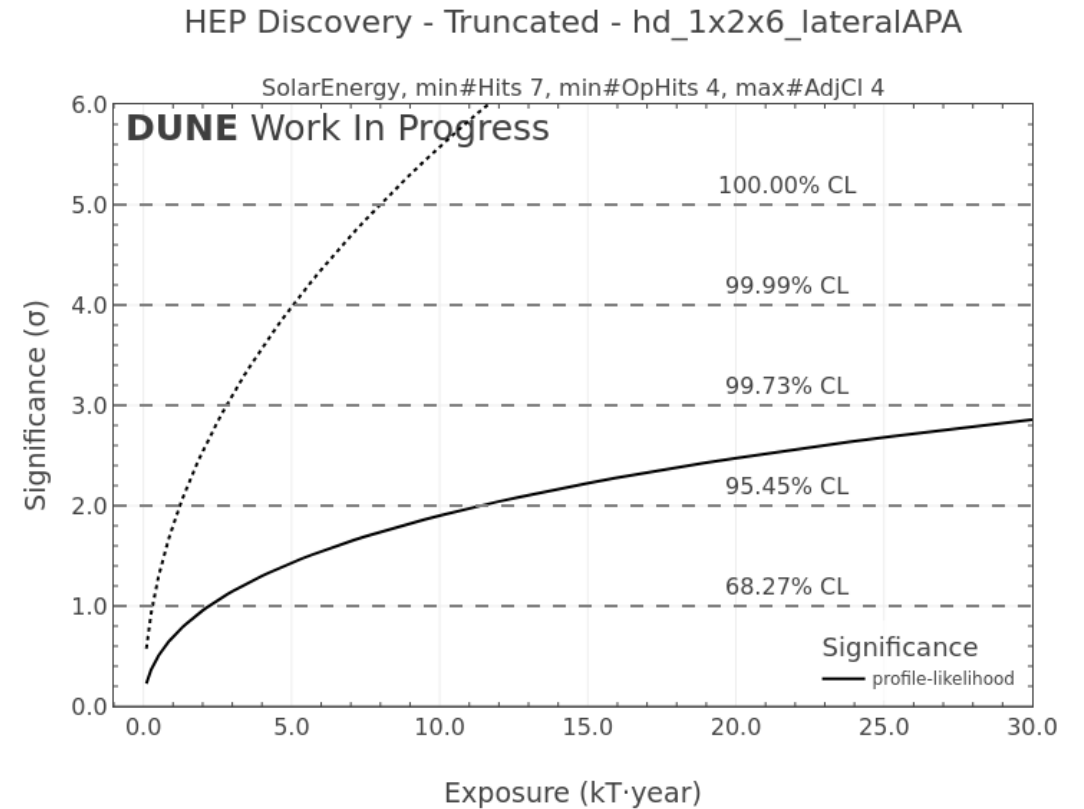
Lower subplot guide: local discovery density, estimated as $z_{\text{local}} / \Delta E$ (sigma per MeV), where z_{local} comes from the per-bin discovery test statistic. Compare where discovery is concentrated

HD Lateral

Significance

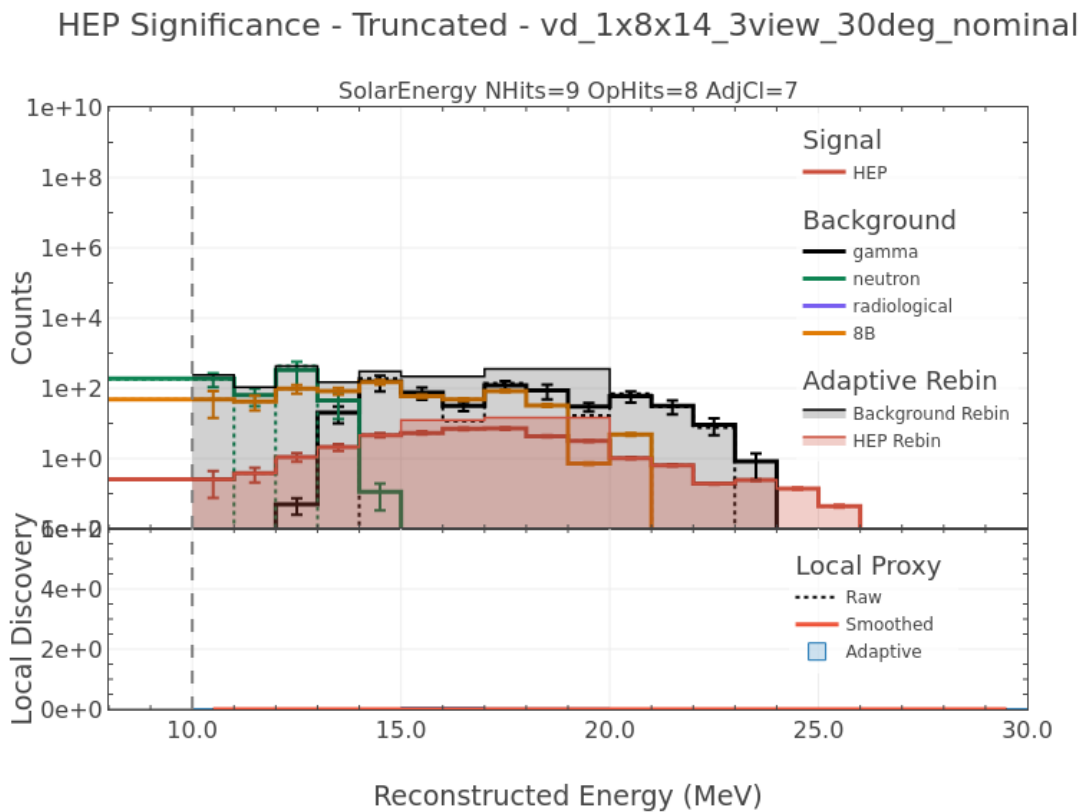


Exposure

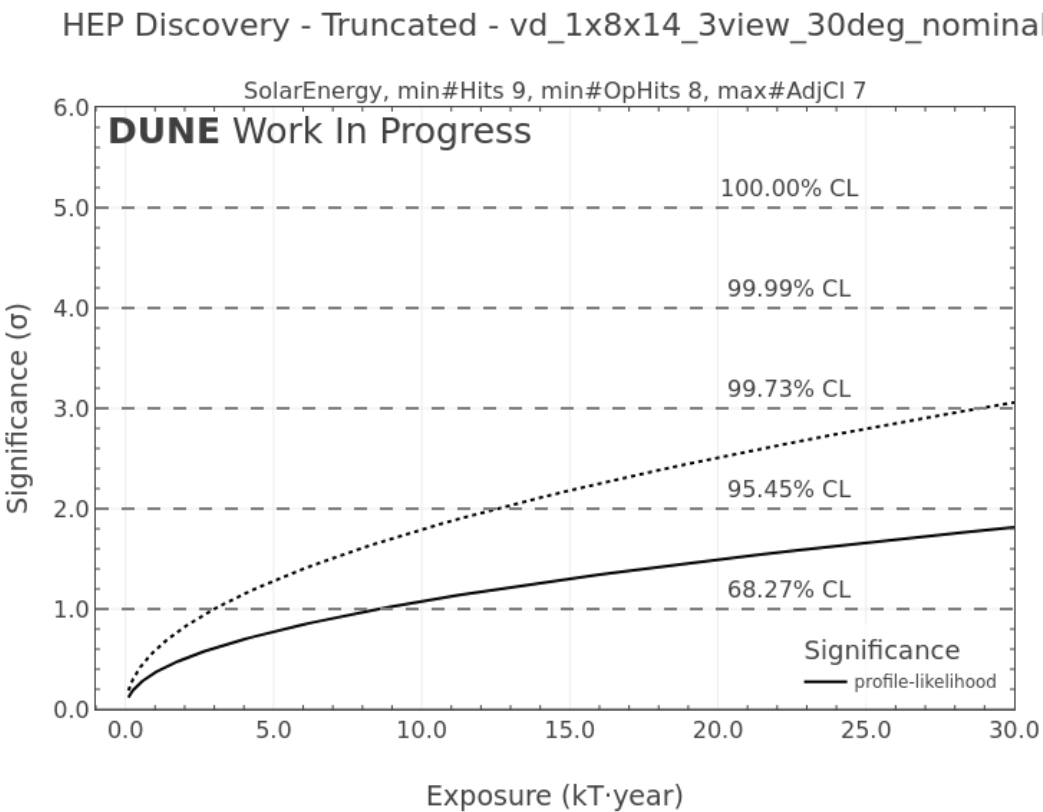


Lower subplot guide: local discovery density, estimated as $z_{\text{local}} / \Delta E$ (sigma per MeV), where z_{local} comes from the per-bin discovery test statistic. Compare where discovery is concentrated

Significance



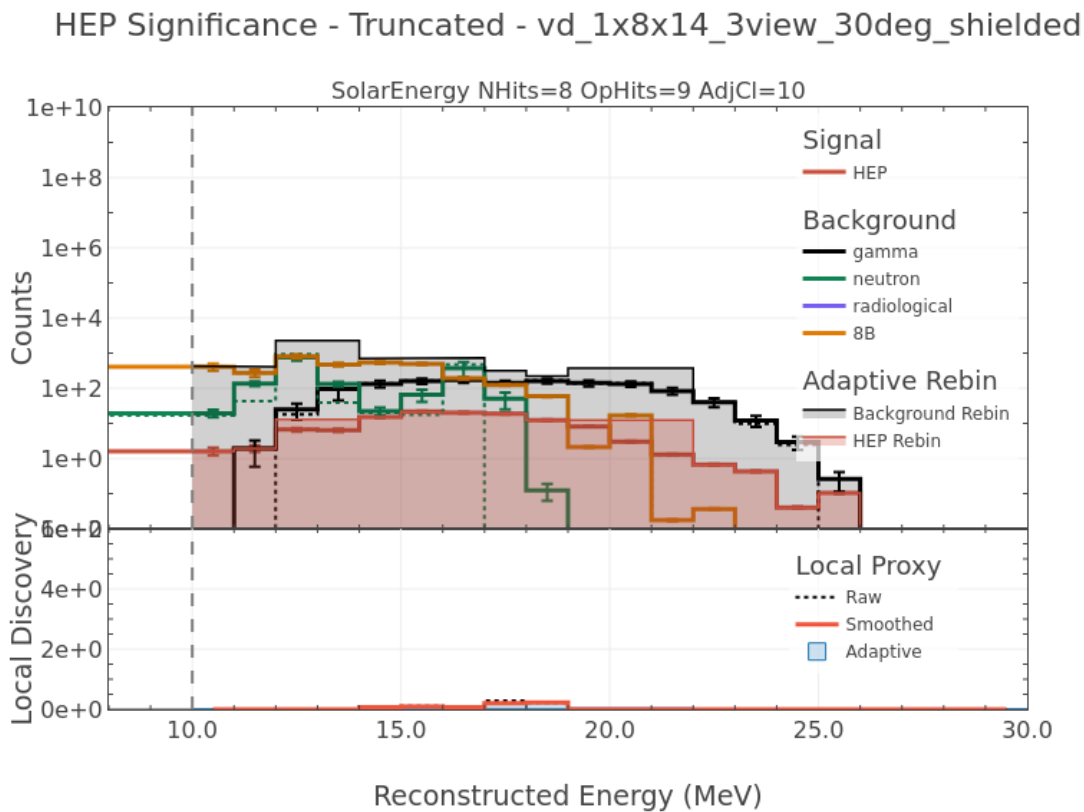
Exposure



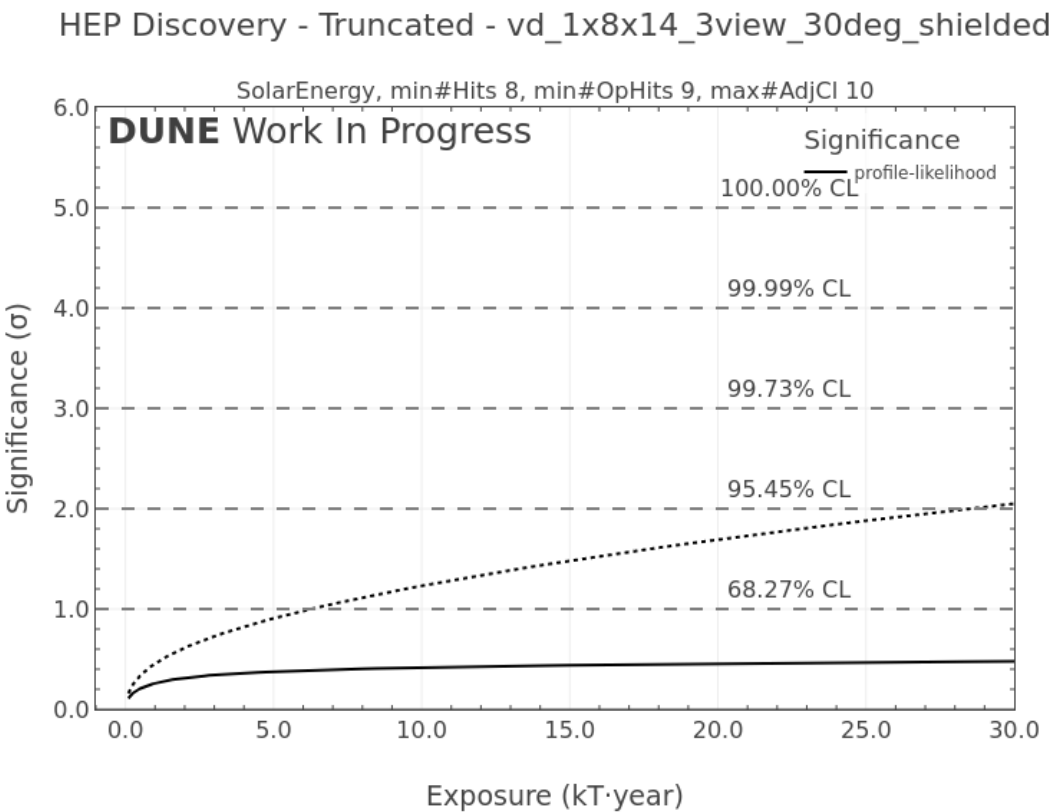
Lower subplot guide: local discovery density, estimated as $z_{\text{local}} / \Delta E$ (sigma per MeV), where z_{local} comes from the per-bin discovery test statistic. Compare where discovery is concentrated

VD Bottom Shielded

Significance



Exposure

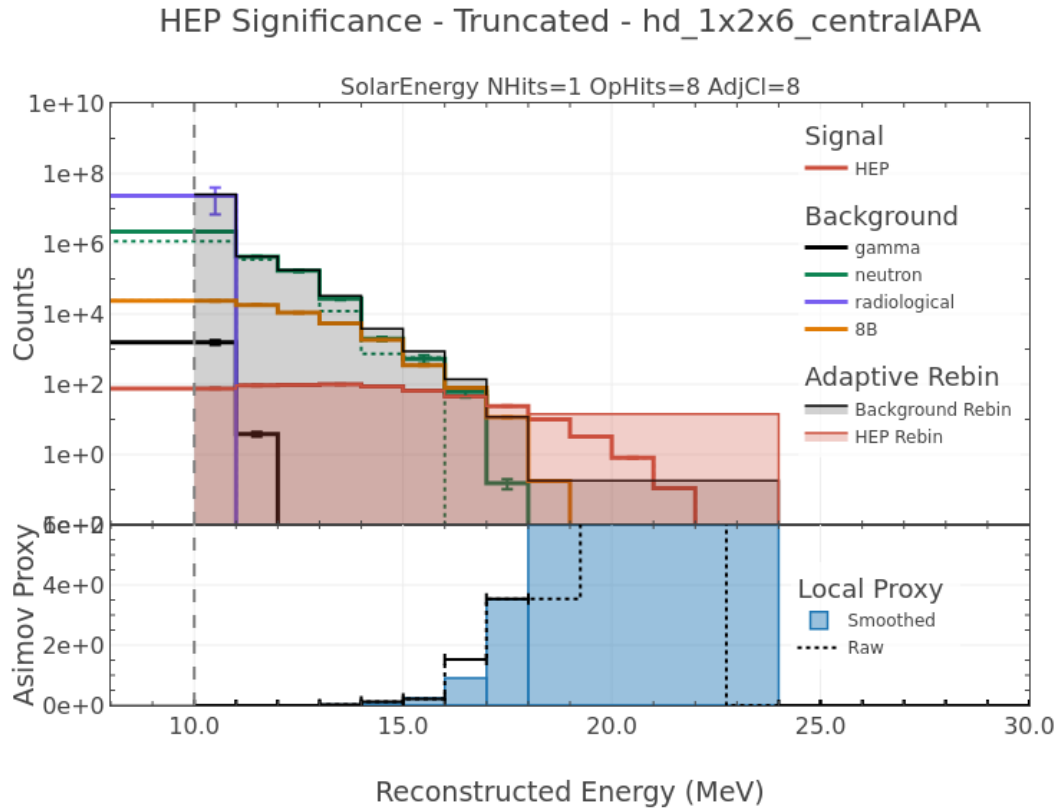


Lower subplot guide: local discovery density, estimated as $z_{\text{local}} / \Delta E$ (sigma per MeV), where z_{local} comes from the per-bin discovery test statistic. Compare where discovery is concentrated

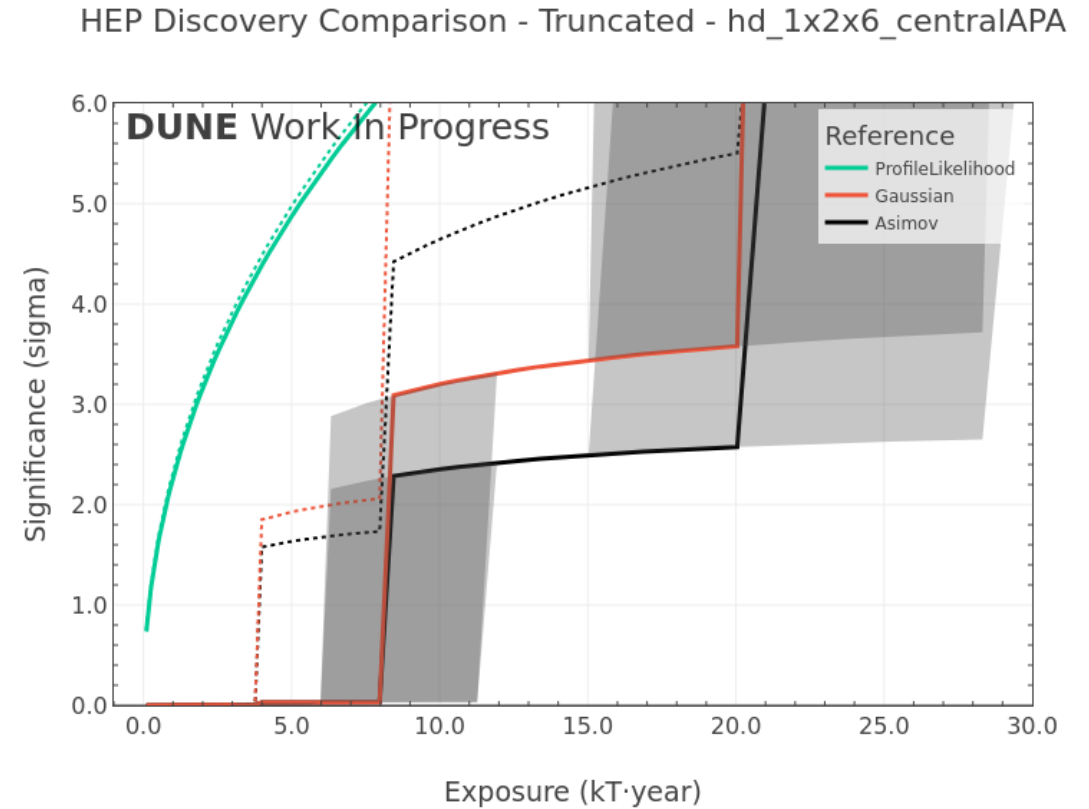
Reference Comparison

HD Central

Significance Reference Comparison

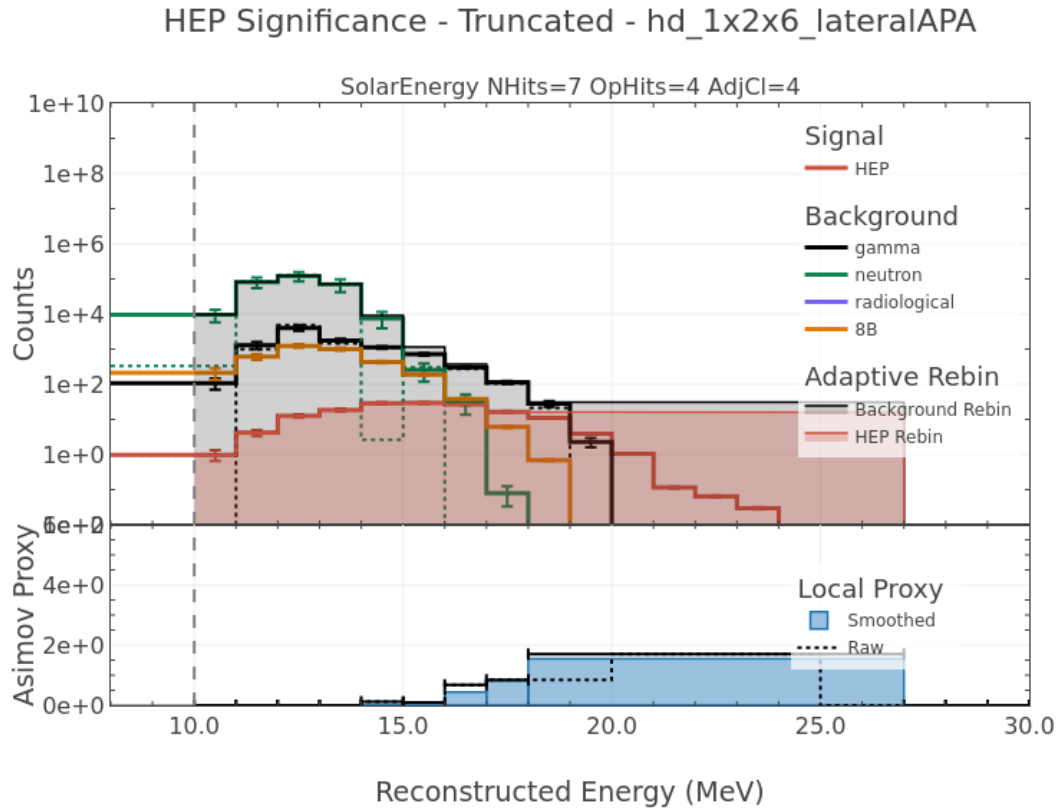


Exposure Reference Comparison

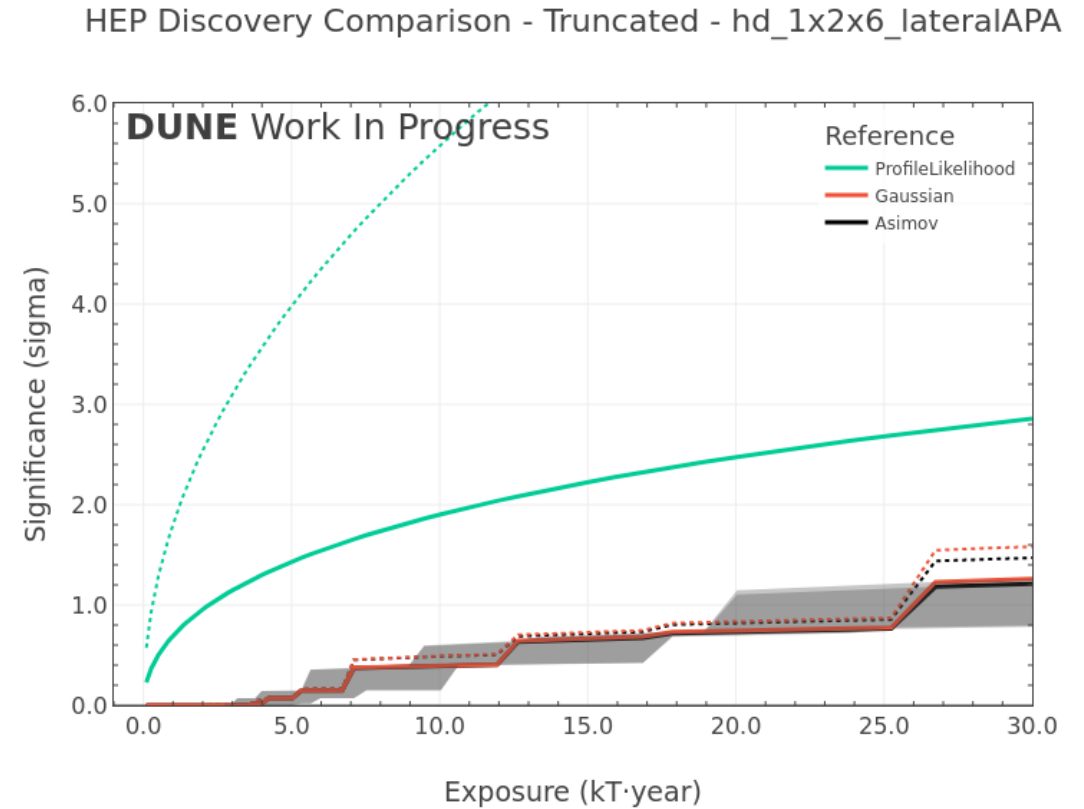


HD Lateral

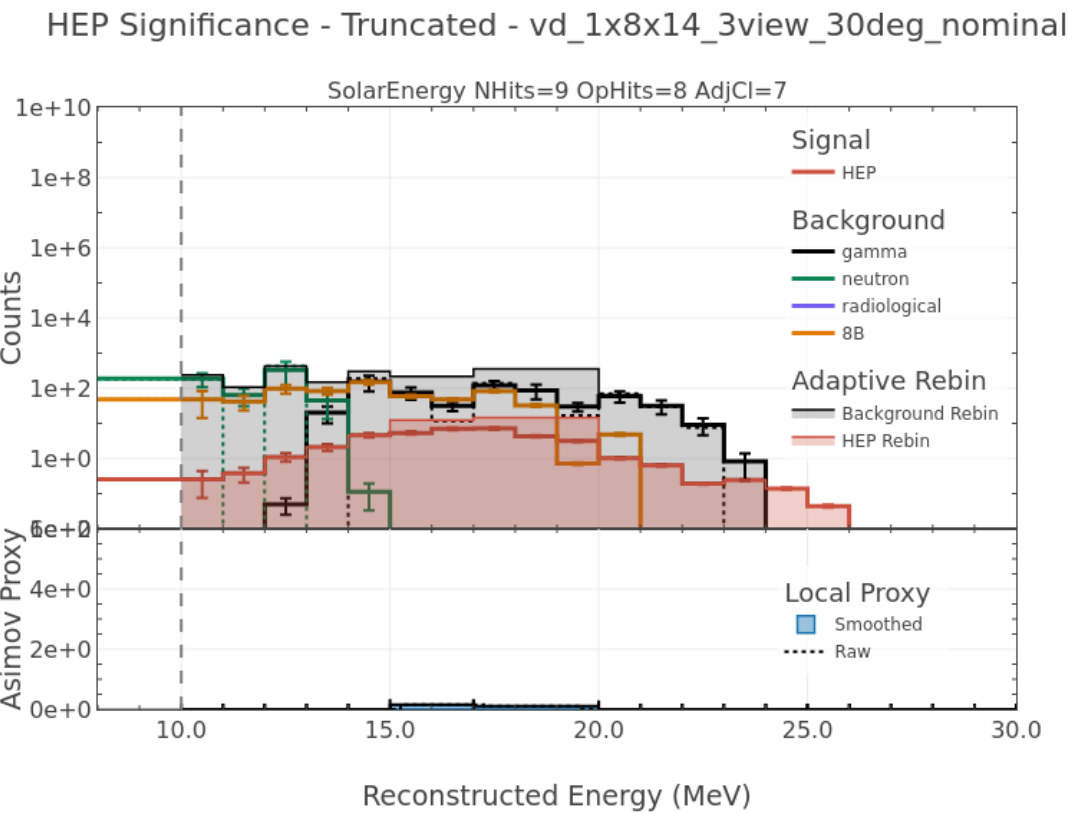
Significance Reference Comparison



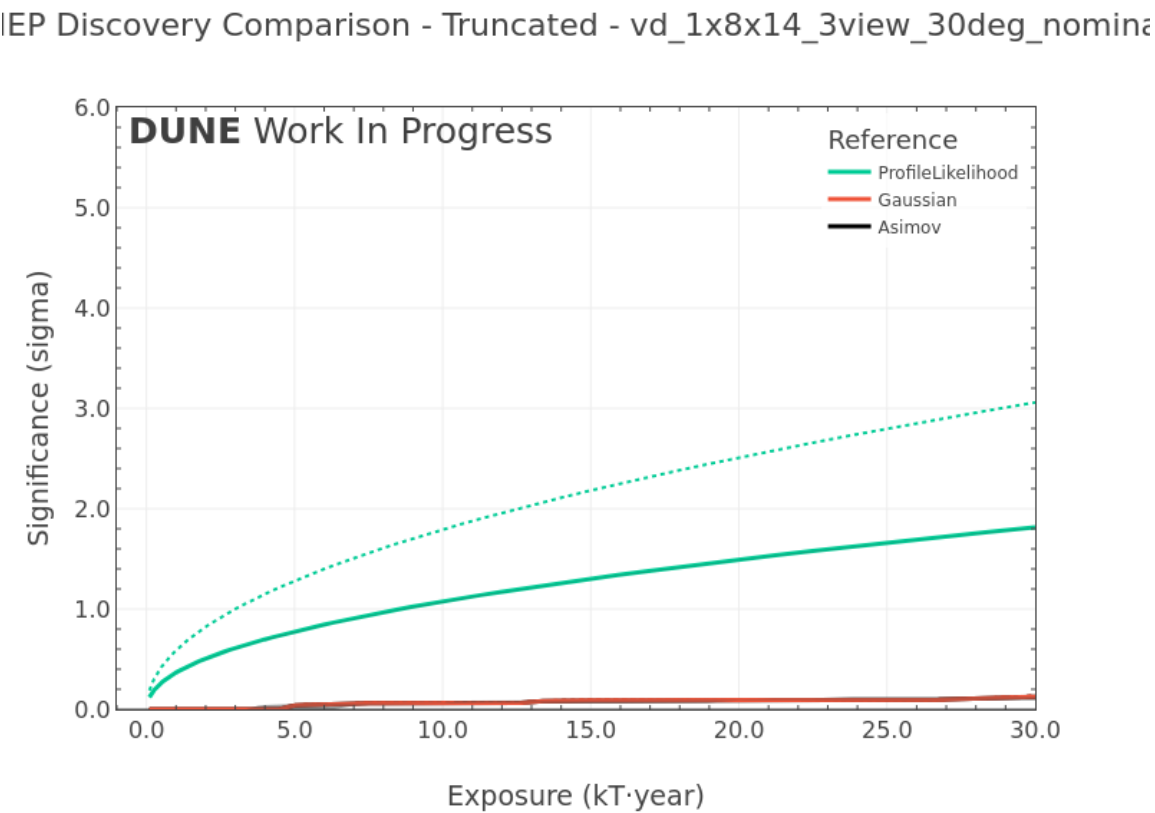
Exposure Reference Comparison



Significance Reference Comparison

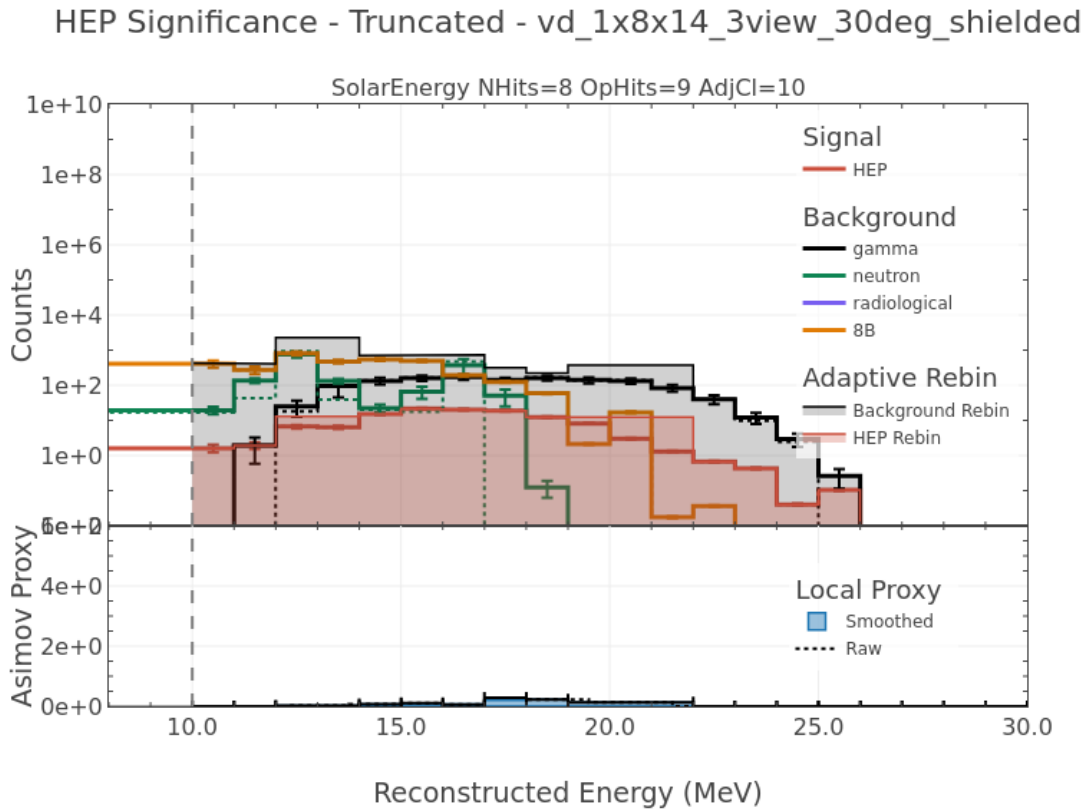


Exposure Reference Comparison

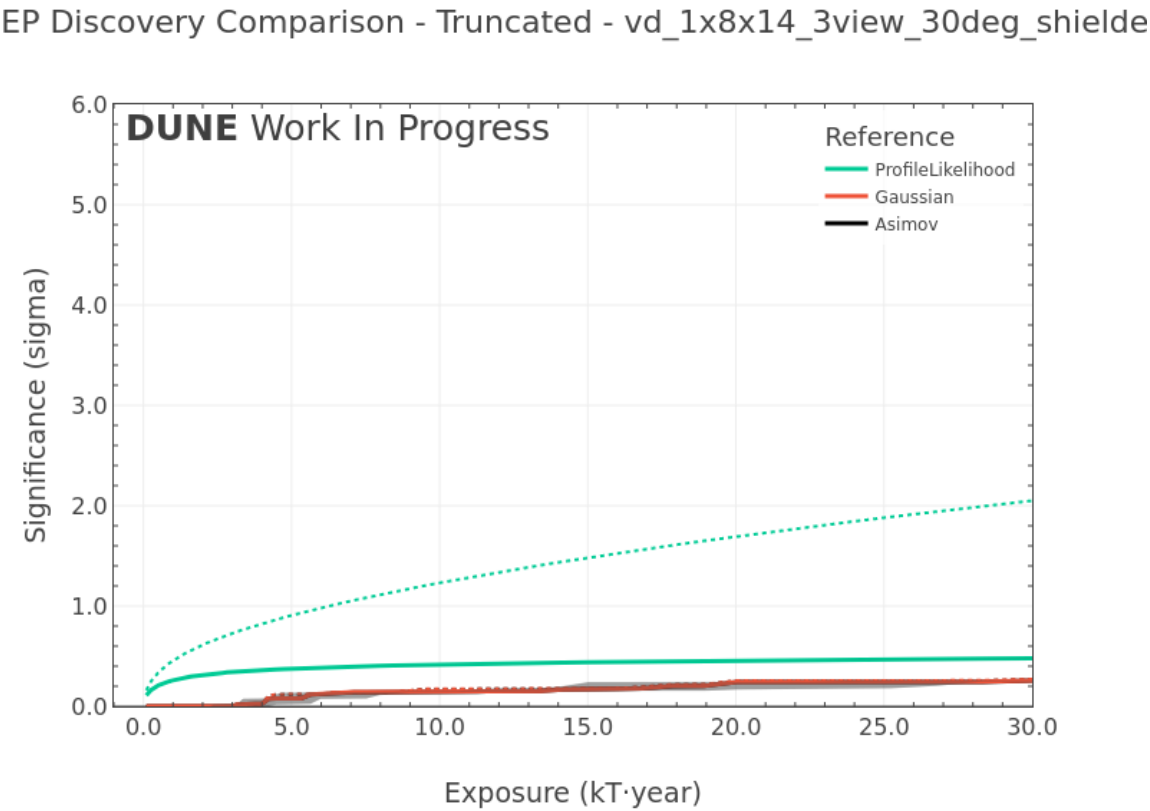


VD Bottom Shielded

Significance Reference Comparison



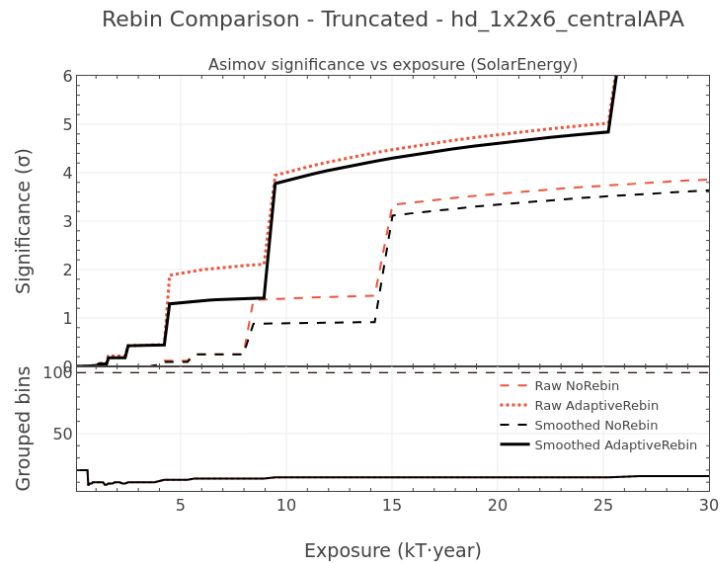
Exposure Reference Comparison



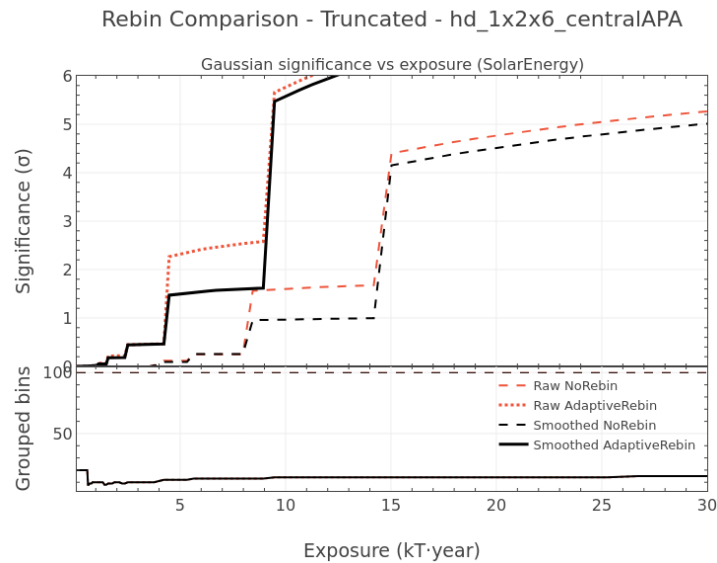
Adaptive Rebin Comparison

HD Central

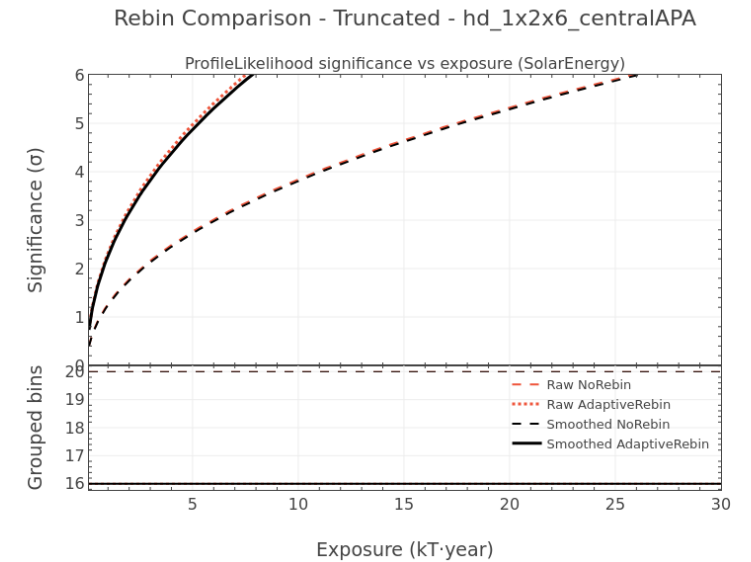
Adaptive Rebin Comparison (Asimov)



Adaptive Rebin Comparison (Gaussian)



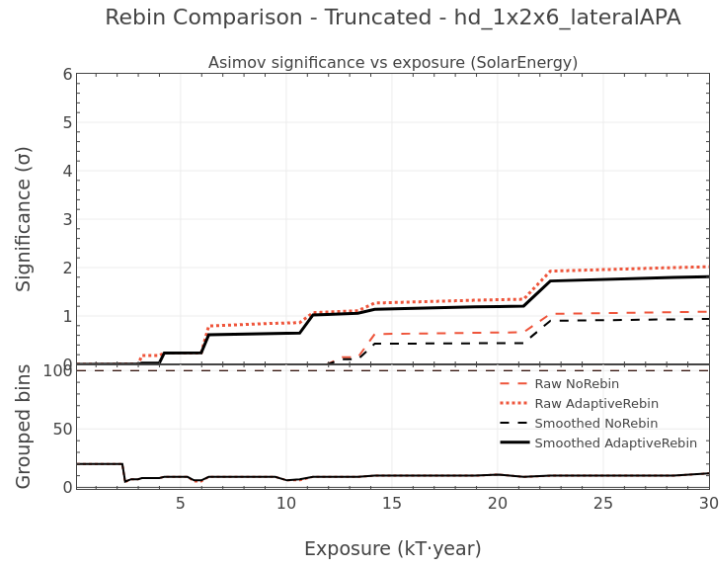
Adaptive Rebin Comparison (ProfileLikelihood)



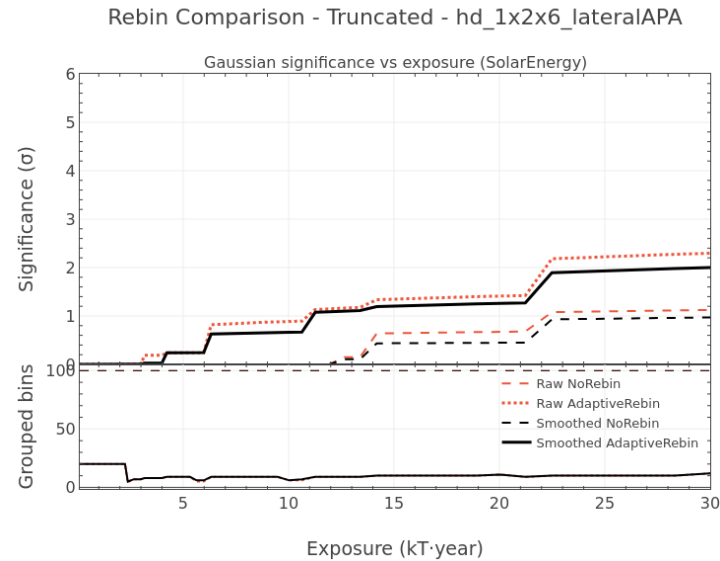
How to compare: Left = Asimov, middle = Gaussian, right = ProfileLikelihood. Compare the exposure turn-on point near threshold 10 MeV, relative ordering between methods, and curve smoothness/step behavior after adaptive rebinning.

HD Lateral

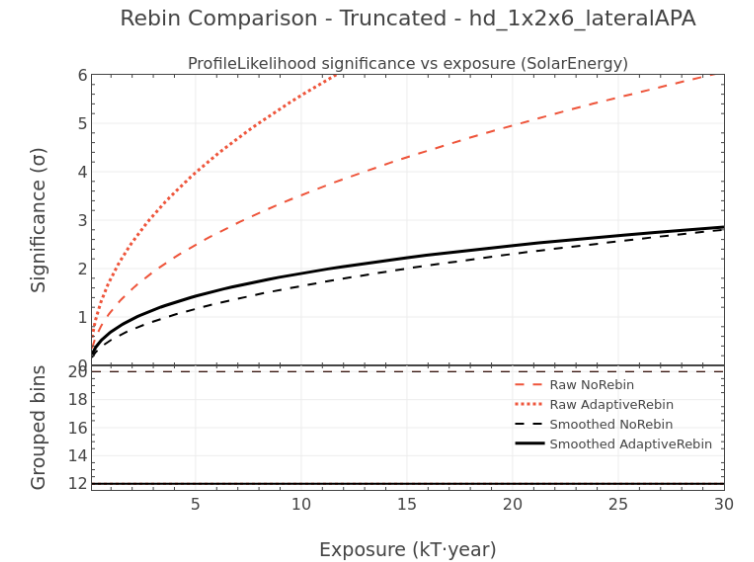
Adaptive Rebin Comparison (Asimov)



Adaptive Rebin Comparison (Gaussian)



Adaptive Rebin Comparison (ProfileLikelihood)

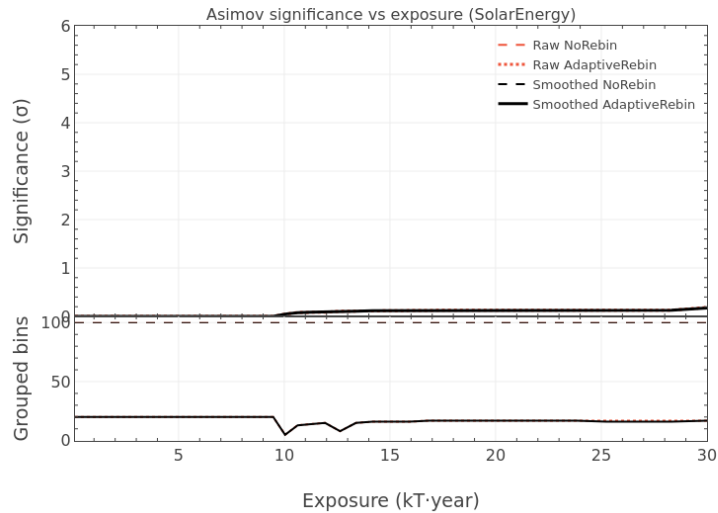


How to compare: Left = Asimov, middle = Gaussian, right = ProfileLikelihood. Compare the exposure turn-on point near threshold 10 MeV, relative ordering between methods, and curve smoothness/step behavior after adaptive rebinning.

VD Top

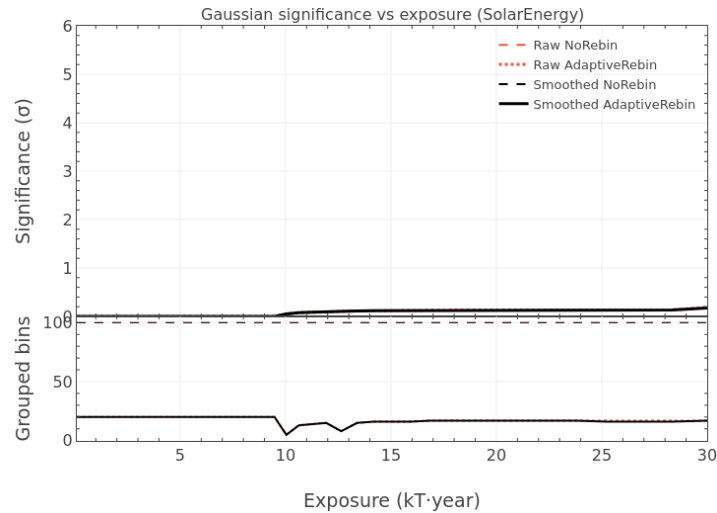
Adaptive Rebin Comparison (Asimov)

Rebin Comparison - Truncated - vd_1x8x14_3view_30deg_nominal



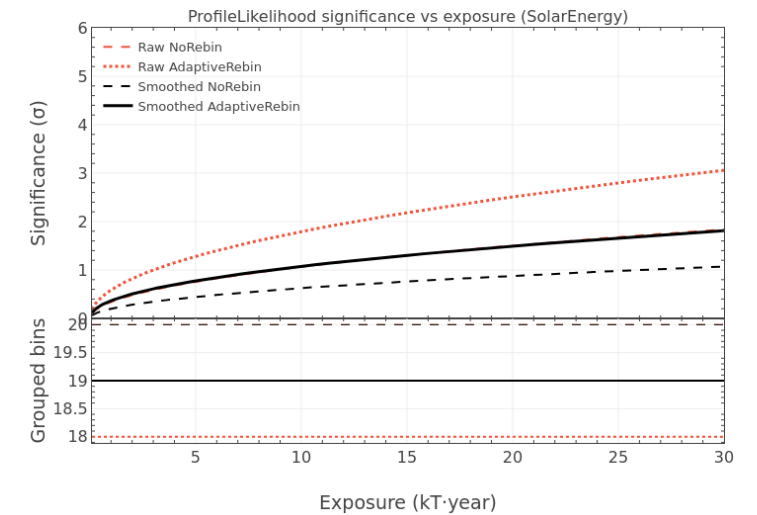
Adaptive Rebin Comparison (Gaussian)

Rebin Comparison - Truncated - vd_1x8x14_3view_30deg_nominal



Adaptive Rebin Comparison (ProfileLikelihood)

Rebin Comparison - Truncated - vd_1x8x14_3view_30deg_nominal

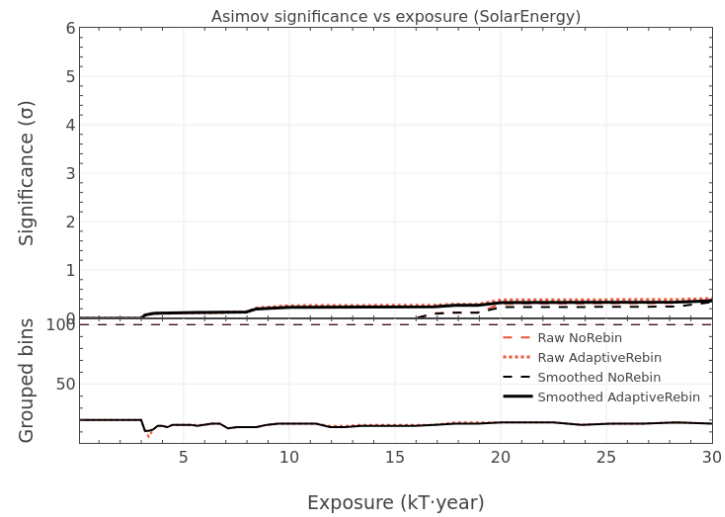


How to compare: Left = Asimov, middle = Gaussian, right = ProfileLikelihood. Compare the exposure turn-on point near threshold 10 MeV, relative ordering between methods, and curve smoothness/step behavior after adaptive rebinning.

VD Bottom Shielded

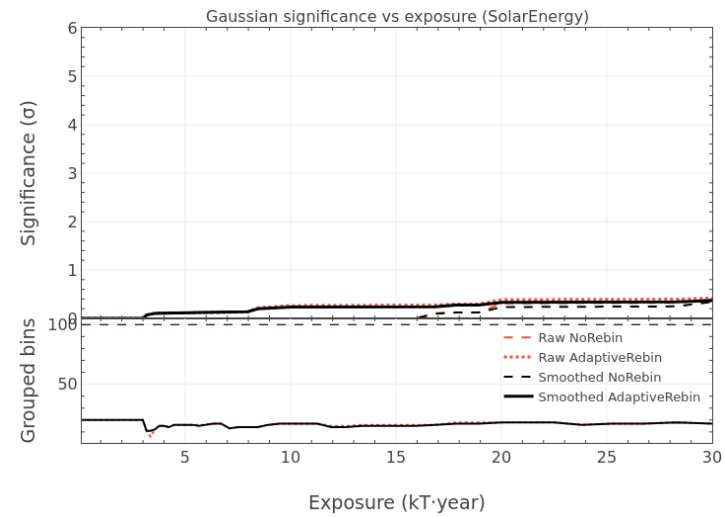
Adaptive Rebin Comparison (Asimov)

Rebin Comparison - Truncated - vd_1x8x14_3view_30deg_shielded



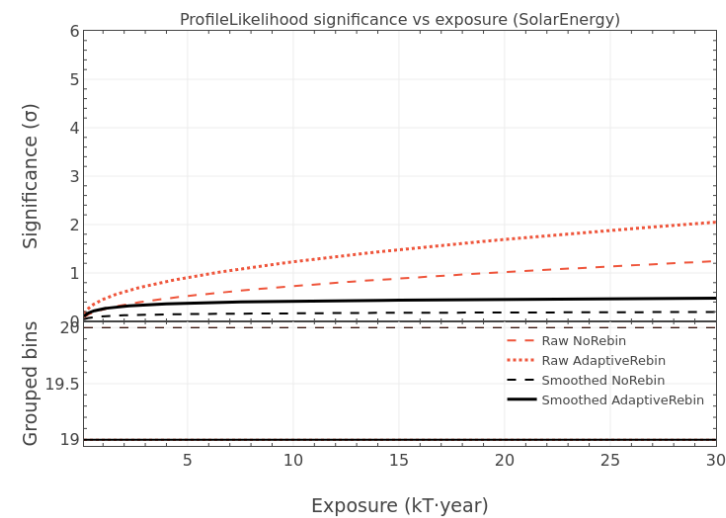
Adaptive Rebin Comparison (Gaussian)

Rebin Comparison - Truncated - vd_1x8x14_3view_30deg_shielded



Adaptive Rebin Comparison (ProfileLikelihood)

Rebin Comparison - Truncated - vd_1x8x14_3view_30deg_shielded



How to compare: Left = Asimov, middle = Gaussian, right = ProfileLikelihood. Compare the exposure turn-on point near threshold 10 MeV, relative ordering between methods, and curve smoothness/step behavior after adaptive rebinning.

Best HEP Cuts by Config

Config	NHits	OpHits	AdjCI	Significance
HD Central	1	8	8	9.843
HD Lateral	7	4	4	1.810
VD Top	9	8	7	0.171
VD Bottom Shielded	8	9	10	0.361

Coverage and Notes

- Config coverage in best-cut JSON outputs:
 - nominal: 4
 - reduced: 4
 - truncated: 4
- Table values are read from workflow-generated JSON at generation time.
- Re-run script to refresh this folder after each workflow run:
- `/usr/bin/python3 scripts/generate_hep_presentation.py --folder truncated`